



Republic of the Philippines
Department of Education
NATIONAL CAPITAL REGION
SCHOOLS DIVISION OF QUEZON CITY
QUEZON CITY SCIENCE HIGH SCHOOL



Golden Acres Road cor. Misamis Street, Bago Bantay, Quezon City

**TIBOK: An nRF52840 XIAO Sced Sense-Based Edge-IoMT Wearable ECG Monitoring
System Using a 1D Convolutional Neural Network (1D-CNN) for Real-Time
Arrhythmia Detection**

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Borja, Blither Jayce
Campo, Aaron Jed E.
Labao, Rey Tristan M.
Luyun, Dan Rui D.
Quintana, Renz Winry M.
Walo, Jaden Zachary V.
Amparo, Raya Kalayaan R.
Delos Santos, Ayeza Leigh F.
Luzentales, Morrie J.
Palmiano, Ema Frances Lyd G.
* Tado, Sophia Lauren A.
Velez, Gabrielle P.
Student Researchers

Quezon City Science High School
Quezon City, National Capital Region

Skjoukje U. Jemoya, PhD
Research Adviser
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RATIONALE

According to the World Health Organization (2025), cardiovascular diseases remain the leading driver of global daily mortality, accounting for an estimated 19.8 million annual deaths. Within developing healthcare networks like the Philippines, the management of these conditions is severely bottlenecked by extreme geographical isolation, structural deficits, and acute health workforce shortages. Data from the Philippine Statistics Authority (2025) reveals that cerebrovascular and ischemic heart diseases represent the second largest cause of death in the country, accounting for over 44,260 annual mortalities. Crucially, paroxysmal arrhythmia acts as an immediate clinical precursor of these fatal cerebrovascular events, elevating a patient's risk of ischemic stroke by 4- to 5-fold (Ceriales et al., 2022). Compounding this biological threat is a severe systemic deficit in medical resources; the Philippine Heart Association (2025) highlights a stark workforce crisis where the country suffers a ratio of only one cardiovascular specialist for every 44,000 Filipinos, failing drastically against the WHO standard of 1 per 25,000. Because paroxysmal arrhythmias are inherently intermittent, transient, and frequently asymptomatic, traditional episodic hospital spot-checks completely fail to capture anomalous cardiac events during everyday activity, exposing a critical public health gap that demands decentralized, continuous ambulatory screening at the primary care level (Ahmed et al., 2026).

To mitigate these diagnostic gaps, contemporary clinical frameworks deploy ambulatory instruments such as continuous 24-to-48-hour Holter monitors, patches, and telemetry nodes (GE Healthcare, 2025). However, widespread deployment of these conventional systems across rural Philippine communities is restricted by severe economic and operational barriers. Standard Holter devices are physically bulky and burdensome for continuous long-term wear, introducing high upfront equipment costs, intensive clinical maintenance, and frequent vulnerabilities to lead

misplacement or detachment which degrades signal integrity (GE Healthcare, 2025). Furthermore, as identified by Flores-Castañeda et al. (2025), existing commercial monitors remain tightly constrained by proprietary data opacity, locking raw physiological data within closed corporate ecosystems and preventing open-source interconnectivity with public health networks. Consequently, low-income patients in remote areas remain entirely isolated from proactive screening, forced instead to rely on a critically congested national infrastructure of only 49 stroke-ready hospitals nationwide to access standard cardiovascular diagnostics (Sese & Guillermo, 2023).

While consumer smartwatches have attempted to bridge this gap, their clinical utility in underserved settings is constrained by cost barriers, closed-source classification algorithms, and high-throughput wireless architectures that drain massive battery reserves during continuous transmission (Kim et al., 2023). To directly resolve these limitations, this study introduces TIBOK: an open-source, capital-efficient, wearable electrocardiogram (ECG) monitoring framework designed explicitly for localized, real-time edge-based arrhythmia detection. To eliminate reliance on expensive cloud servers and unstable rural internet connections, the proposed system utilizes the nRF52840 XIAO Seeed Sense microcontroller to execute low-power, continuous data acquisition directly on the wrist-worn device. Rather than masking diagnostic metrics behind proprietary software, the framework combines a traditional on-chip digital signal processing (DSP) pipeline for baseline filtering with an optimized, 8-bit integer-quantized (INT8) One-Dimensional Convolutional Neural Network (1D-CNN) running locally via TensorFlow Lite for Microcontrollers.

This specific combination of a 1D-CNN architecture and the nRF52840 creates an ideal technological synergy for resource-constrained environments. By leveraging the chip's native

ARM Cortex-M4 hardware Floating-Point Unit (FPU), the system can handle complex mathematical tasks with a microscopic electrical footprint, preserving battery longevity while executing real-time local inference directly from low-noise biopotential dry-skin electrodes (Ahmed et al., 2023). Crucially, the system bypasses wireless data bottlenecks by compressing calculated R-R intervals and model classification flags into tightly structured, serialized byte arrays streamed via Bluetooth Low Energy (BLE 5.4) with Data Length Extension (DLE) to a cross-platform Flutter mobile application. By transforming these continuous telemetry packets into immediate visual diagnostic tools—specifically real-time time-series tachograms and non-linear Poincaré scatter plots—the platform equips local community health workers with unmistakable visual signatures of transient arrhythmias (Gruwez et al., 2021). Evaluated against the clinically annotated validation standards of the MIT-BIH Arrhythmia Database, this project serves as a highly accessible, equitable, and actionable digital screening tool tailored specifically for primary care environments across marginalized Philippine populations.

This project aligns with United Nations Sustainable Development Goals (SDGs) 3, 9, 10, and 17 to ensure public welfare-aligned development. By deploying a capital-efficient edge-screening framework that catches intermittent arrhythmias before they escalate into lethal strokes, the device actively advances SDG 3 (Good Health and Well-Being) within underserved communities (World Health Organization, 2025). The creation of an optimized, resource-constrained TinyML biomedical tool furthers SDG 9 (Industry, Innovation, and Infrastructure) by proving that advanced artificial intelligence can be tailored for low-cost, decentralized public health infrastructure (International Telecommunication Union, 2024). Furthermore, by engineering an open-source alternative that bypasses the steep financial and geographical barriers associated with urban medical institutions, the system supports SDG 10

(Reduced Inequalities), democratizing diagnostic access across differing socioeconomic backgrounds. Finally, the development of this framework bridges academic research with public health needs, advancing the localized collaborative mandates of SDG 17 (Partnerships for the Goals).

From a national perspective, this study directly conforms to the long-term vision of AmBisyon Natin 2040 through the pillars of Malasakit, Pagbabago, and Kaunlaran. By extending continuous, clinical-grade cardiac surveillance to geographically isolated and marginalized populations, the project promotes Malasakit through healthcare equity. It drives Pagbabago by reducing the extreme financial vulnerability of rural families cut off from urban medical networks (National Economic and Development Authority, 2023). Concurrently, it fuels Kaunlaran by positioning the Philippines as an active innovator in localized, embedded TinyML biomedical designs. This approach translates directly into the operational directives of the DOST Harmonized National R&D Agenda (HNRDA). Specifically, it maps onto the Health Sector through its development of low-cost, accessible telehealth diagnostics, while simultaneously advancing the Industry, Energy, and Emerging Technology Sector by pioneering practical, low-power artificial intelligence applications engineered for real-world public health environments (Department of Science and Technology, 2022).

Review of Related Literature

Global and Local Burden of Cardiovascular Disease

Cardiovascular disease continues to be one of the most pressing public health concerns in the world. The World Health Organization (2025) identifies cardiovascular diseases as the leading cause of death globally, with an estimated 19.8 million deaths in 2022 alone. What makes this situation even more alarming is that a large proportion of these deaths occur in low- and middle-income countries, where access to preventive care, timely diagnosis, and specialized treatment is often limited. In practical terms, this means that many individuals who are already at risk do not receive the kind of monitoring that could prevent complications or support earlier intervention.

The burden of cardiovascular disease is not only medical but also social and economic. In many resource-limited settings, patients often delay consultation because of cost, distance from health facilities, or lack of available specialists. When health systems cannot provide regular screening and follow-up, cardiovascular conditions are often discovered only after symptoms have progressed. This late detection increases the likelihood of hospitalization, disability, and even death, while also placing greater financial strain on families and communities. The World Bank likewise emphasized that low- and middle-income countries continue to face a disproportionate share of the noncommunicable disease burden, while primary care systems remain underdeveloped for early cardiovascular management.

This challenge is also visible in the Philippine context. The country continues to face structural limitations in delivering timely cardiovascular care, particularly in rural and underserved communities. Shortages in trained health personnel, uneven distribution of services, and limited access to specialized cardiac centers make early screening difficult for many patients.

As a result, there is a continuing need for low-cost technologies that can support cardiac monitoring outside large hospitals. The literature suggests that decentralized, affordable, and user-friendly monitoring devices could help address this gap by bringing screening closer to the community setting.

Evidence from other low-resource countries supports this direction. Armand et al. (2023) developed a low-cost wireless cardiovascular monitoring system in Cameroon using an IoT-based platform to measure blood pressure and heart rate. Their results showed that the system was accurate, practical, and usable for patient monitoring in low-cost environments. Similarly, Shah et al. (2021) found that portable 12-lead ECG screening in primary health centers could be clinically useful and economically viable for high-risk individuals. These studies show that affordable cardiovascular technologies can strengthen early detection efforts where conventional care is difficult to access. They also reinforce the idea that innovation in cardiac care should not be limited to high-income settings.

Limitations of Current Wearable ECG Systems

Wearable ECG technology has emerged as a promising solution for continuous heart monitoring. Unlike traditional hospital-based testing, wearables can capture data during daily activity, which makes them especially useful for detecting intermittent arrhythmias that may not appear during short clinical examinations. Flores-Castañeda et al. (2025) observed that wearable systems have advanced considerably through the integration of passive sensors, predictive algorithms, and digital platforms for remote patient monitoring. These developments show that wearable health technology is moving toward more proactive and continuous care rather than episodic assessment alone.

Despite these advances, the literature also makes clear that wearable ECG systems still face important limitations. Flores-Castañeda et al. (2025) noted that methodological heterogeneity, limited validation against clinical standards, and unresolved concerns about interoperability, privacy, and regulation continue to limit reproducible clinical benefit. In other words, while the technology is promising, it still cannot fully replace clinical-grade monitoring unless its reliability is clearly established. For wearable devices to be trusted in screening settings, they must be tested under conditions that reflect real-world use and compared against accepted medical references.

Commercial smartwatches have brought wearable ECG screening into everyday life, but their performance is still uneven. The Fitbit Heart Study demonstrated that consumer wearables can detect atrial fibrillation in large populations and may help identify undiagnosed cases earlier. However, later studies showed that performance varies by device and that many readings remain inconclusive or require confirmation by a medical ECG. Barrera et al. (2025) likewise concluded that smartwatches have excellent screening potential but should not be used as standalone diagnostic tools. This distinction matters because screening and diagnosis are not the same thing; a wearable may be useful for prompting further evaluation, but it must still be held to rigorous standards before it can support clinical decisions independently.

Although wearable ECG systems can support continuous monitoring, intermittent arrhythmias are often difficult to capture in a way that clearly links symptoms with ECG findings. Given the transient and often asymptomatic nature of paroxysmal arrhythmia, patient-activated event recording has an established role in improving diagnostic yield when symptoms and cardiac rhythm findings do not align in time. Rajanna et al. (2022) reviewed external cardiac loop recorders and their role in capturing symptom-rhythm correlation, noting

that patient-triggered recording enables clinically useful correlation between subjective symptoms and objective rhythm findings during infrequent arrhythmic episodes that continuous monitoring alone may not adequately contextualize. Consistent with this, Pistelli et al. (2025) reported real-world diagnostic yield data from loop recorder deployment, reinforcing the clinical value of event-triggered recording as a complement to continuous ambulatory monitoring. These findings support the inclusion of a patient-activated event marker within the TIBOK system to enable direct correlation between wearer-reported symptoms and recorded ECG findings.

Another persistent issue is signal quality. Wearable ECG devices are designed for use during movement, but motion itself introduces noise that can reduce accuracy. Studies by Al-Sheikh (2023) and An et al. (2022) show that adaptive filtering and motion-aware preprocessing can reduce artifact contamination and improve ECG quality during exercise or everyday activity. These findings suggest that wearable ECG systems should be designed not only to detect rhythm abnormalities but also to preserve signal integrity under realistic usage conditions.

The hardware interface is equally important. Goyal et al. (2022) showed that dry electrodes can support continuous biopotential monitoring, but their performance is strongly influenced by skin-contact impedance, electrode material, and hydration level. Their findings showed that titanium electrodes performed better than stainless steel, particularly in dry-skin conditions. This is significant because it demonstrates that accuracy is affected not just by software or algorithms, but also by the physical materials used to acquire the signal. For wearable ECG devices, hardware design is therefore a central factor in overall reliability.

Embedded and On-Device Arrhythmia Detection

The literature increasingly supports the use of low-cost embedded systems for health monitoring, provided that the sensing, communication, and processing pipeline is carefully designed. Filgueiras et al. (2025) showed that affordable wearable hardware can still achieve acceptable reliability for continuous vital-sign monitoring. This is important because it challenges the assumption that effective monitoring must rely on expensive platforms. Instead, the evidence suggests that low-cost embedded systems can still provide meaningful clinical support when their components are chosen and integrated wisely.

Wireless communication is another essential part of wearable design. Serrani et al. (2024) found that BLE synchronization and data transmission significantly affect the accuracy and efficiency of a wearable multi-sensor platform. This supports the use of Bluetooth Low Energy in systems intended for compact, low-power applications. In particular, it strengthens the rationale for a BLE-capable board such as the nRF52840 Sense, which can support local sensing while maintaining communication efficiency. For a wearable intended for continuous use, this combination of low power and stable transmission is highly valuable.

On the analytical side, machine learning offers a practical way to interpret sequential physiological data. Yi et al. (2023) showed that CNN-based models, particularly when combined with pruning and quantization for lightweight deployment, are effective for recognizing patterns in wearable sensor streams without requiring hand-crafted feature extraction which is an approach directly relevant to the INT8-quantized architecture used in this study. This is relevant to ECG analysis because arrhythmias often appear as time-dependent patterns rather than simple isolated values. CNNs are well suited for learning these patterns directly from the signal, making them a strong option for embedded classification.

Related studies also support CNN-based processing in John et al. (2026) suggest that convolutional models can effectively combine wearable sensor inputs, which is useful in systems that may integrate ECG with motion or other physiological signals. In a device that must operate in real time, this capability becomes especially useful because the model can process multiple inputs together instead of treating them separately. As wearable health systems become more complex, such integrated analysis will likely become more important.

More recent research has strengthened the case for on-device inference through TinyML and edge computing. Hasan et al. (2026) emphasized that local processing is advantageous when privacy, speed, and energy efficiency matter. Hizem et al. (2025) also showed that ECG anomaly detection can be implemented on resource-limited hardware, including Arduino and Raspberry Pi platforms. Kim et al. (2023) demonstrated that a TinyML-based ECG system can work effectively on low-power devices using public datasets and embedded inference. These findings support the use of an INT8-quantized 1D-CNN on the nRF52840 Sense rather than depending on cloud processing, which would consume more power and introduce communication delays.

Public ECG databases further strengthen the feasibility of embedded arrhythmia detection. Open-source datasets such as MIT-BIH and PTB-XL make it possible to train and validate models in a transparent and reproducible way. Studies using these databases have shown that embedded CNN-based ECG classification can achieve strong results in low-resource implementations. Guhdar et al. (2025) also reported high performance on MIT-BIH and PTB-XL, reinforcing the idea that accurate ECG classification is possible even with compact hardware. Taken together, these studies show that embedded arrhythmia detection is not only technically possible but increasingly practical for low-cost wearables.

Diagnostic Validation and Performance Metrics

A wearable ECG system cannot be considered useful unless its performance is validated against a recognized standard. Shreffler and Huecker (2025) explain that diagnostic testing accuracy is commonly assessed using sensitivity, specificity, predictive values, and accuracy. These measures are important because they indicate whether a test is identifying the correct cases and avoiding false alarms. In a screening context, this distinction is critical because the value of the device depends on how well it can guide the next clinical step.

Sensitivity measures how well a device detects true cases, while specificity measures how well it correctly identifies those without the condition. Predictive values are equally important because they help interpret the practical meaning of a positive or negative result in a specific population. For wearable ECG systems, these measures determine whether the output is sufficiently reliable to support follow-up care. A device that produces too many false positives may create unnecessary anxiety and referrals, while one with too many false negatives may miss patients who actually need treatment.

If a validation sample does not reflect the true population distribution, then PPV and NPV may not fully represent real-world performance. Shreffler and Huecker (2025) explain that prevalence affects predictive values, and wearable validation studies should therefore be interpreted in light of population context. This is particularly relevant to wearable validation studies, where samples are often selected for convenience or clinical access rather than population representativeness. For that reason, comparing the wearable output with a cardiologist-interpreted 12-lead ECG remains the most appropriate reference approach.

This kind of validation does more than confirm technical performance, it establishes clinical credibility. A wearable ECG system may perform well in a controlled dataset, but if it

cannot align reasonably with expert interpretation, its practical value remains limited. Validation is therefore not a final step added after development; it is central to proving that the system can support real screening use.

Visual Support for Arrhythmia Interpretation

Beyond automated classification, visualization can help make rhythm interpretation more intuitive. Tachograms and Poincaré plots are useful because they present RR-interval behavior in a way that highlights rhythm irregularity more clearly than raw waveform data alone. Flanders et al. (2022) showed that time-selected tachogram and Poincaré plot windowing can reveal transient cardiac rhythm changes that may be missed in short waveform segments. This matters because arrhythmias are often intermittent, and a brief strip of ECG may not always capture the abnormal pattern.

Gruwez et al. (2021) further demonstrated the value of these visual tools in arrhythmia interpretation. In the INTERPRET-AF study, physicians performed better when tachogram and Poincaré plot information was added to waveform review. This suggests that geometric visualization can improve diagnostic clarity by making subtle rhythm patterns more recognizable. In practical terms, such tools can support both medical staff and non-specialist users by presenting rhythm information in a more interpretable form.

Recent work has extended the use of Poincaré-based visualization to address challenges more specific to wearable and automated contexts. Yasuoka et al. (2025) addressed a problem particularly relevant to continuous wearable ECG recording, raw Poincaré plots can be distorted by noisy or failed signal segments common in ambulatory Holter-style monitoring, and their proposed buffered Poincaré plot method, which filters out inconsistent RR-intervals before visualization, improved automatic atrial fibrillation diagnostic accuracy by approximately 8%

over conventional methods. This is directly relevant to a wearable system such as TIBOK, where motion artifacts and intermittent signal quality are expected challenges. Beyond supporting human interpretation, Poincaré geometry has also been shown to support automated detection: Rao et al. (2023) demonstrated its use in algorithmic atrial fibrillation detection, while Aldughayfiq et al. (2023) similarly combined Poincaré representation of ECG signals with deep-learning-based image classifiers for cardiovascular disease detection. Together, these findings suggest that geometric visualization holds value across both clinician-facing interpretation and algorithmic diagnostic pathways.

For a mobile-based wearable system, visualization provides a useful layer of support beyond algorithmic classification. Users may not always be trained to read ECG traces, but a tachogram or Poincaré plot can make irregularity easier to notice. This is especially useful in screening contexts, where the goal is not to replace clinical interpretation but to help identify when follow-up may be needed. In that sense, visualization increases the practical value of the wearable by making the output easier to understand.

Synthesis

The reviewed literature shows a strong common concern with the burden of cardiovascular disease and the need for more accessible ways to detect and monitor heart-related conditions. One clear similarity across the studies is the recognition that cardiovascular disease remains the leading cause of death worldwide and continues to place a disproportionate burden on low- and middle-income countries (World Health Organization, 2025; World Heart Federation, 2023). The World Bank (2025) likewise emphasized that these countries continue to experience a high burden of noncommunicable diseases while primary healthcare systems remain limited in providing early cardiovascular care. In the Philippine setting, recent literature

also highlights continuing challenges such as shortages of trained health professionals, unequal access to specialized cardiac services, and limited opportunities for early screening. Because of these conditions, many of the studies point toward the importance of affordable and decentralized technologies that can support cardiovascular monitoring outside major hospitals. Studies by Armand et al. (2023) and Shah et al. (2021) further show that low-cost cardiovascular monitoring systems and portable ECG screening can provide practical support for early detection in resource-limited communities.

A second major similarity across the studies is the growing interest in wearable ECG systems as tools for continuous cardiac monitoring. Several works show that wearable devices can help identify atrial fibrillation and other rhythm abnormalities during daily activities, making them useful for preliminary screening and long-term observation (Lubitz et al., 2021; Mannhart et al., 2023; Barrera et al., 2025). Because many arrhythmias occur only occasionally, studies also show that patient-activated event recording can improve the correlation between symptoms and ECG findings when continuous monitoring alone does not capture the episode (Rajanna et al., 2022; Pistelli et al., 2025). However, the same studies consistently emphasize that wearable ECG devices still have important limitations. Many require confirmation using a clinical ECG, while concerns about validation, algorithm transparency, interoperability, and regulatory compliance continue to limit their use as standalone diagnostic tools (Flores-Castañeda et al., 2025; Barrera et al., 2025). This creates an important contrast in the literature. Wearable ECG devices have considerable screening potential, but their current role remains primarily supportive rather than fully diagnostic.

The studies also differ in how they explain the limitations of wearable ECG systems. Some focus on clinical validation and diagnostic reliability, while others concentrate on signal

acquisition and hardware performance. On the clinical side, several papers emphasize that wearable ECG systems should be compared with physician-interpreted 12-lead ECG to establish their usefulness in screening and diagnosis (Shreffler & Huecker, 2025; Podaru & David, 2021; Yuniadi et al., 2024; Chan et al., 2022). On the technical side, other studies show that motion artifacts, electrode material, and skin-electrode impedance continue to affect signal quality during ambulatory monitoring (Goyal et al., 2022; An et al., 2022; Al-Sheikh et al., 2023). Although these studies examine different aspects of wearable ECG systems, they all suggest that both hardware quality and proper clinical validation are necessary to produce reliable results.

The literature also shows a clear similarity in the direction of current solutions. Many researchers are moving toward embedded, low-power, and on-device processing for wearable health systems. Several studies explain that TinyML and edge computing make local ECG analysis more practical by reducing power consumption, improving privacy, and minimizing dependence on cloud connectivity (Kim et al., 2023; Hasan et al., 2026; Hizem et al., 2025). Other studies show that CNN-based models are effective in learning rhythm patterns directly from ECG signals while remaining suitable for lightweight embedded deployment (Yi et al., 2023; John et al., 2026; Filgueiras et al., 2025). The difference among these studies lies mainly in their implementation, with some emphasizing model optimization, others focusing on embedded hardware, and others examining communication efficiency through Bluetooth Low Energy. Even so, they all support the same general idea that embedded processing can make arrhythmia detection more practical for continuous monitoring in low-resource environments.

Another important similarity is the emphasis on making ECG results easier to interpret. Some studies focus on automated classification using machine learning, while others emphasize visual tools such as tachograms and Poincaré plots (Flanders et al., 2022; Gruwez et al., 2021).

These approaches differ in method, but they share the same objective of making rhythm abnormalities easier to recognize. More recent studies also show that improved Poincaré plot techniques can reduce the effects of noisy ECG recordings and improve atrial fibrillation detection, while other researchers demonstrate that these visual representations can also support deep learning models (Yasuoka et al., 2025; Rao et al., 2023; Aldughayfiq et al., 2023). Together, these findings suggest that wearable ECG systems can benefit from combining automated analysis with visual support, particularly in screening settings where users may not have specialized ECG knowledge.

Validation remains one of the most consistent themes in the literature. The studies agree that sensitivity, specificity, predictive values, and accuracy are essential for determining whether a wearable ECG system can support clinical decision-making (Shreffler & Huecker, 2025). At the same time, they also point out an important difference between technical performance and real-world usefulness. Predictive values depend on disease prevalence, meaning that results obtained from selected validation samples may not fully represent performance in the general population (Shreffler & Huecker, 2025; Yuniadi et al., 2024). This distinction is important because a wearable device may achieve high performance during testing but still produce different results when used in community screening. For this reason, the literature consistently supports validation against recognized clinical standards before wearable ECG systems can be considered reliable for practical use.

Overall, the literature points in the same general direction even though the studies differ in focus and method. The shared message is that cardiovascular disease remains a major public health concern, wearable ECG systems have significant potential for improving early detection, and embedded machine learning technologies can make continuous monitoring more accessible

and practical. At the same time, the differences across the studies show that important challenges remain, including signal quality, hardware performance, algorithm transparency, and clinical validation. These similarities and differences help define the current state of research and show where the present study fits within it. Rather than presenting a complete solution, the literature demonstrates an ongoing effort to develop wearable ECG systems that are more accurate, more accessible, and more suitable for continuous monitoring in everyday settings.

RESEARCH QUESTIONS

1. What is the system accuracy level of TIBOK and its companion Flutter mobile application in real-time arrhythmia detection and visualization, based on its successful end-to-end system integration?
2. What are the significant differences between the:
 - 2.1. F1 score of TIBOK and FP32 reference model in terms of detection performance across test recordings?
 - 2.2. End-to-end processing latency of TIBOK and FP32 reference model from signal acquisition to result visualization?
 - 2.3. Specificity of TIBOK and FP32 reference model monitors in correctly identifying non-arrhythmia ECG segments?
3. In adult patients:
 - 3.1 What are the sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of TIBOK for detecting arrhythmia, using a GE MAC 5500 12-lead ECG interpreted by a cardiologist as the reference standard?
 - 3.2 What is the level of agreement between TIBOK and the cardiologist-interpreted GE MAC 5500 12-lead ECG findings in classifying arrhythmia versus non-arrhythmia, as measured by Cohen's kappa (κ)?

HYPOTHESES

H_{01} : The system accuracy level of TIBOK and its companion Flutter mobile application is less than or equal to a 95% accuracy in real-time arrhythmia detection and visualization, based on its successful end-to-end system integration.

H_{11} : The system accuracy level of TIBOK and its companion Flutter mobile application has a 95% or greater accuracy in real-time arrhythmia detection and visualization, based on its successful end-to-end system integration.

H_{02} : There is no significant difference in the F1-score between the TIBOK system and the FP32 reference model in detecting arrhythmia across test recordings.

H_{12} : There is a significant difference in the F1-score between the TIBOK system and the FP32 reference model in detecting arrhythmia across test recordings.

H_{03} : There is no significant difference in the end-to-end processing latency between the TIBOK system and the FP32 reference model.

H_{13} : There is a significant difference in the end-to-end processing latency between the TIBOK system and the FP32 reference model.

H_{04} : There is no significant difference in the specificity between the TIBOK system and the FP32 reference model in identifying non-arrhythmia ECG segments.

H_{14} : There is a significant difference in the specificity between the TIBOK system and the FP32 reference model in identifying non-arrhythmia ECG segments.

H_{05} : The sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of TIBOK for detecting arrhythmia, using a GE MAC 5500 12-lead ECG as interpreted by a cardiologist is less than or equal to the acceptable clinical standard.

H_{15} : The sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) of TIBOK for detecting arrhythmia, using a GE MAC 5500 12-lead ECG as interpreted by a cardiologist is greater than the acceptable clinical standard.

H_{06} : There is no level of agreement between TIBOK and the cardiologist-interpreted GE MAC 5500 12-lead ECG in classifying arrhythmia versus non-arrhythmia ($\kappa \leq 0$).

H_{16} : There is a statistically significant level of agreement between TIBOK and the cardiologist-interpreted GE MAC 5500 12-lead ECG in classifying arrhythmia versus non-arrhythmia ($\kappa > 0$).

GENERAL OBJECTIVE

This study aims to develop and evaluate a low-cost, open-source, and real-time wearable cardiovascular monitoring system that tracks ECG signals, detects arrhythmia, and visualizes cardiac data via a mobile-accessible Flutter application using the MIT-BIH Arrhythmia Database for performance validation.

SPECIFIC OBJECTIVES

1. To evaluate the system accuracy level of the TIBOK system and its companion Flutter mobile application for real-time arrhythmia detection and visualization following end-to-end integration.
2. To compare the F1 score of the TIBOK system against the desktop FP32 baseline using the MIT-BIH Arrhythmia Database to determine if a significant difference exists in detection performance.
3. To compare the end-to-end processing latency of the TIBOK system against the desktop FP32 baseline under a fixed 360 Hz sampling rate to determine if a significant difference exists.
4. To evaluate and compare the specificity of the TIBOK system against the desktop FP32 baseline using the MIT-BIH Arrhythmia Database to determine if a significant difference exists in identifying non-arrhythmia segments.
5. To evaluate how TIBOK detects arrhythmia in adult participants compared to a cardiologist-interpreted GE MAC 5500 12-lead ECG (or comparator wearable, where unavailable) reference standard, by determining its sensitivity, specificity, PPV, and NPV as pilot-level descriptive estimates, and measuring the level of agreement in heart rate (Bland-Altman) and rhythm classification via Cohen's kappa (κ), within a purposively

selected proof-of-concept sample of 10 adult participants (5 with a history of arrhythmia, 5 without).

EXPECTED OUTCOMES

This study is expected to determine whether the TIBOK wearable ECG monitoring system and its companion Flutter mobile application can successfully achieve end-to-end integration for continuous cardiovascular monitoring. Specifically, the study will evaluate whether the system can continuously acquire ECG signals, perform on-device arrhythmia classification using the embedded INT8 One-Dimensional Convolutional Neural Network (1D-CNN), transmit processed data through Bluetooth Low Energy (BLE), and provide real-time visualization of ECG waveforms, heart rate, tachograms, Poincaré plots, and arrhythmia classification results while achieving the system accuracy threshold specified in Hypothesis H11.

The study is also expected to determine whether quantizing the arrhythmia detection model from FP32 to INT8 affects diagnostic performance and computational efficiency. Through the evaluation of Hypotheses H12 to H14, the embedded INT8 model will be compared with the FP32 desktop reference model using the MIT-BIH Arrhythmia Database in terms of F1-score, specificity, and end-to-end processing latency. The results will establish whether the embedded implementation can preserve diagnostic performance while providing the computational efficiency required for real-time edge deployment, or whether quantization introduces performance trade-offs that require further refinement of the model or optimization process.

Furthermore, the study is expected to determine the clinical screening capability of TIBOK through a proof-of-concept human study against a cardiologist-interpreted GE MAC 5500 12-lead ECG reference standard (or comparator wearable device where the reference standard is unavailable for a given session). Specifically, the study will evaluate Hypotheses H15 and H16

by determining descriptive sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) estimates, and whether TIBOK's rhythm classifications demonstrate agreement with the reference standard as measured by Cohen's kappa (κ), alongside heart rate agreement assessed via Bland-Altman analysis. Given the purposive proof-of-concept sample of 10 participants (5 with a history of arrhythmia, 5 without), these results will be treated as hypothesis-generating pilot evidence rather than generalizable diagnostic accuracy, intended to guide future studies with larger and more representative populations.

Overall, the findings from both the technical validation and the pilot clinical validation are expected to determine whether TIBOK demonstrates sufficient technical performance and diagnostic capability to function as a low-cost, open-source wearable ECG screening system for resource-limited settings. Alternatively, the results may indicate that certain aspects of the system, including model quantization, embedded implementation, signal acquisition quality, hardware design, or population generalizability, require further improvement before broader clinical application can be supported

ENGINEERING GOALS

The main engineering goals of this study are to design, build, and create a cost-efficient, accessible, portable cardiorespiratory monitoring system with the ability to continuously acquire and process electrocardiograph (ECG) and photoplethysmography (PPG) signals in real time. The device will include a printed circuit board (PCB) as a base, an nRF52840 XIAO Seeed Sense microcontroller will be integrated, an AD8232 ECG sensor, a MAX30102 pulse oximeter, an MLT9650 magnetic buzzer for audible alert notifications, a 3.7V LiPo Battery for a portable power source, a TP4056 Battery charging module that protects the battery from overheating and overcharging, a Low-Dropout (LDO) voltage regulator that is capable of supplying 3.3V to the nRF52840 XIAO Seeed Sense and the sensors, three pieces of Ag/AgCl ECG Electrodes as a disposable material attached to the body to collect ECG signals, and a 3.5 mm jack cable which connects the electrodes to the AD8232 sensor. The device will be used to gather physiological data, detect potential cardiac abnormalities such as arrhythmia through an embedded signal-processing 1-D Convolutional Neural Network, and the data collected will be transmitted to a mobile application. Flutter (version 3.44.4) is the programming application and Dart (version 3.12.2) will be the programming language used for the codes of the app.

The process will begin by assembling the device and integrating the sensors such as the ECG, PPG, and motion sensors. The ECG model will be prepared for classification; it will be optimized and loaded into the nRF52840 microcontroller. Then, the system will be paired with the Flutter application through Bluetooth Low Energy (BLE) to acquire real-time physiological data and arrhythmia results that will be displayed on the mobile application.

To evaluate, the embedded INT8 model and the FP32 reference model will both process and compare the ECG recordings that came from the MIT-BIH Arrhythmia Database. The

system will also log elapsed time from signal acquisition, and it will record successful acquisition, inference, transmission, and visualization to determine the end-to-end integration rate from the displayed result.

According to Nurrohman (2025), they concluded that MAX30102 is accurate in measuring ECG and SpO₂, with accuracy reaching 96.97% for heart rate and 99.76% for oxygen saturation. Also, according to Tutuko et al. (2021) in a similar study, they detected Atrial Fibrillation (AFib) using a convolutional network, and it reached 98.94% accuracy, 98.97% sensitivity, and 98.97% specificity. According to Medical Devices Safety and Performance (2024), medical electrical equipment should comply with the IEC 60601 standard, which is a requirement for certification in many countries. The unit cost of materials is estimated at ₱3,802, while the estimated cost of the equipment is ₱15,434.00, excluding initial laboratory testing fees.

In addition, this study aims to design and implement a user-friendly interface that is capable of visualizing ECG waveforms, tachograms, and Poincaré plots to enhance the detection of arrhythmia. The process will prioritize optimizing system accuracy, reliability, and real-time performance while maintaining affordability and accessibility. This project aims to provide a practical biomedical monitoring solution as an assistance to conventional cardiovascular monitoring devices.

RESEARCH DESIGN

This study will utilize a quantitative, experimental-comparative research design to develop and evaluate the performance of a wearable ECG monitoring system, TIBOK, which will deploy an 8-bit integer quantized (INT8) one-dimensional convolutional neural network (1D-CNN) on a Seeed Studio XIAO nRF52840 Sense microcontroller. The embedded wearable model will be compared with a 32-bit floating-point (FP32) desktop reference model to determine whether quantization and edge deployment will affect diagnostic performance and processing efficiency. The independent variable will be the execution platform, specifically the wearable edge device and the desktop baseline. The dependent variables will include F1-score, specificity, and end-to-end processing latency. The model architecture, ECG segment length, and preprocessing procedure will be held constant across both platforms to ensure fair comparison.

To ensure clinical and experimental relevance, the study will use two validation tracks. First, the wearable model will be evaluated using the MIT-BIH Arrhythmia Database, which contains annotated ECG records that will serve as the benchmark dataset for comparing TIBOK with the FP32 baseline. Second, the prototype will undergo a proof-of-concept human study involving 10 purposively selected adult participants: 5 with a documented history of arrhythmia and 5 without known arrhythmia. For participants whose arrhythmia presents intermittently rather than continuously, multiple or extended recording sessions will be conducted to increase the likelihood of capturing an arrhythmic episode within the observation window. Participants with arrhythmia will undergo multiple or extended recording sessions to increase the likelihood of capturing intermittent arrhythmic episodes within the observation window. Each participant will undergo simultaneous recording using TIBOK and a reference device; the GE MAC 5500 12-lead ECG interpreted by a cardiologist, or a comparator wearable (e.g., Apple Watch) where

GE MAC 5500 access is not feasible for a given session. The reference reading will be used to calculate descriptive sensitivity, specificity, positive predictive value, negative predictive value, and Cohen's kappa (κ) for rhythm classification agreement, alongside Bland-Altman analysis (mean bias and limits of agreement) for heart rate agreement. Following each recording session, participants will complete a short usability questionnaire (8–10 items, adapted from the System Usability Scale) assessing comfort, wearability, usability, and trust in the TIBOK device and companion application. Given the small purposive sample, statistical treatment will be descriptive only, with results reported as pilot-level, hypothesis-generating evidence rather than inferential diagnostic accuracy. The study will measure diagnostic and agreement outputs through confusion-matrix and Bland-Altman analysis, and will record processing latency using embedded hardware timers. The results will be used to determine whether the wearable INT8 model can provide diagnostic performance comparable to the desktop FP32 reference model while maintaining the real-time efficiency required for wearable arrhythmia screening.

PROCESS FLOWCHART

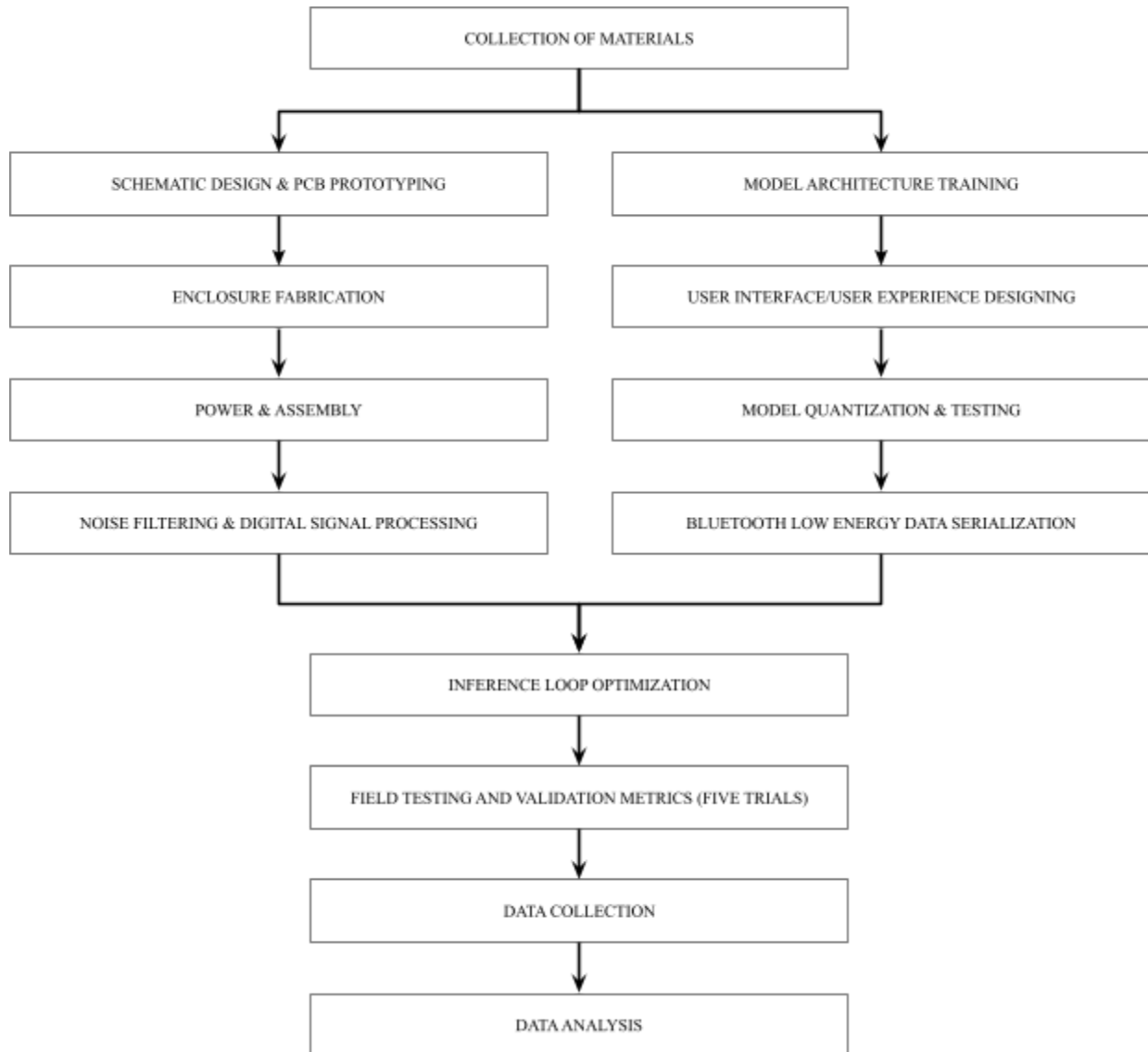


Figure 1

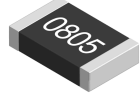
Proposed Process Flowchart for TIBOK.

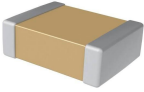
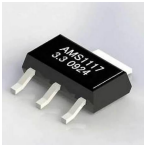
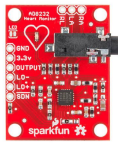
PROCEDURES

A. COLLECTION OF MATERIALS

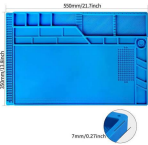
Table 1.1

Materials and equipment to be used.

ITEM #	PIC	DESC.	UNIT(S)	UNIT COST	T. Cost
MATERIALS					
1		nRF52840 XIAO Seeed Sense	1	₱1,500.00	₱1,500.00
2		3.7V 300mAh LiPo Battery	1	₱150.00	₱150.00
3		TP4056 Battery Charger Module	1	₱20.00	₱20.00
4		SMD Resistors Thick Film (0805, 5%, 1/8W, 22Ω, pack of 50)	1	₱45.00	₱45.00

5		Zener Diodes (SOD-123, 5.1V, pack of 20)	1	₱100.00	₱100.00
6		SMD Multilayer Ceramic (MLCC, 0.1uF, 0805, 50V, pack of 50 pcs)	1	₱45.00	₱45.00
7		1N4148WS Switching Diode (SOD-323, pack of 50 pcs)	1	₱50.00	₱50.00
8		SMD Transistors MMBT3904 NPN (SOT-23, pack of 50 pcs)	1	₱70.00	₱70.00
9		AMS1117-3.3 Linear Regulator (SOT-223, pack of 10)	1	₱55.00	₱55.00
10		AD8232 ECG Kit with Module	1	₱500.00	₱500.00

11		MAX 30102 Pulse Oximeter	1	₱175.00	₱175.00
12		Rosin core soldering lead (63/37, 100g)	1	₱200.00	₱200.00
13		Heat Shrink Tubing	1	₱110.00	₱110.00
14		5 x 10 cm Copper Clad Laminate Boards (pack of 5 pcs)	1	₱103.00	₱103.00
15		Jadeway Hyper PETG Filament	1	₱499.00	₱499.00
16		Omron B3U-1000P	1	₱50	₱50
17		Nexperia PESD3V3L4UG TVS	1	₱80	₱80

		diode array, quad-channel ESD protection (SOT143B)			
					TOTAL: ₱3,802
EQUIPMENT					
18		Flux Pen	1	₱115.00	₱115.00
19		Heat-resistant silicone mat (for soldering)	1	₱375.00	₱375.00
20		Wire Cutters (Micro Flush Cutters)	1	₱85.00	₱85.00
21		Anti-Static ESD Wrist Strap	1	₱70.00	₱70.00
21		Ferric Chloride ($FeCl_3$) Etching Solution (38% solution, 500ml)	1	₱150.00	₱150.00

22		PCB Drill Bit Set (0.7mm to 1.2mm Micro Bits, 10 pc set)	1	₱170.00	₱170.00
23		Soldering Iron (Temperature-controll ed)	1	₱825.00	₱825.00
24		Desoldering Pump	1	₱105.00	₱105.00
25		Heat Gun	1	₱200.00	₱200.00
26		Desktop Fume Extractor (with carbon filter)	1	₱1,225.00	₱1,225.00
27		Digital Multimeter (Auto-ranging entry-level)	1	₱450.00	₱450.00

27		Alcohol Prep Pads (box of 100 pcs)	1	₱140.00	₱140.00
28		Nitrile Gloves (box of 100 pcs)	1	₱325.00	₱325.00
31		Creality Ender-3 V3	1	₱11,199.00	₱11,199.00
					TOTAL: ₱15,434.00
GRAND TOTAL					₱19,236.00

PROJECT DESIGN

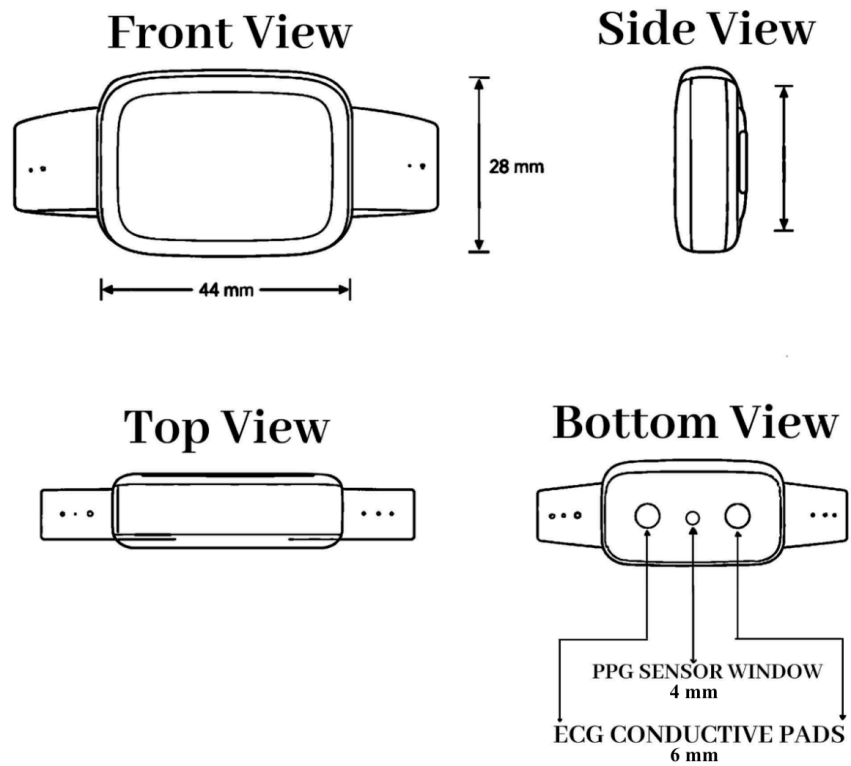


Figure 2

2D Front View, Side View, and Underside View Render of Proposed Project Design for TIBOK Device.

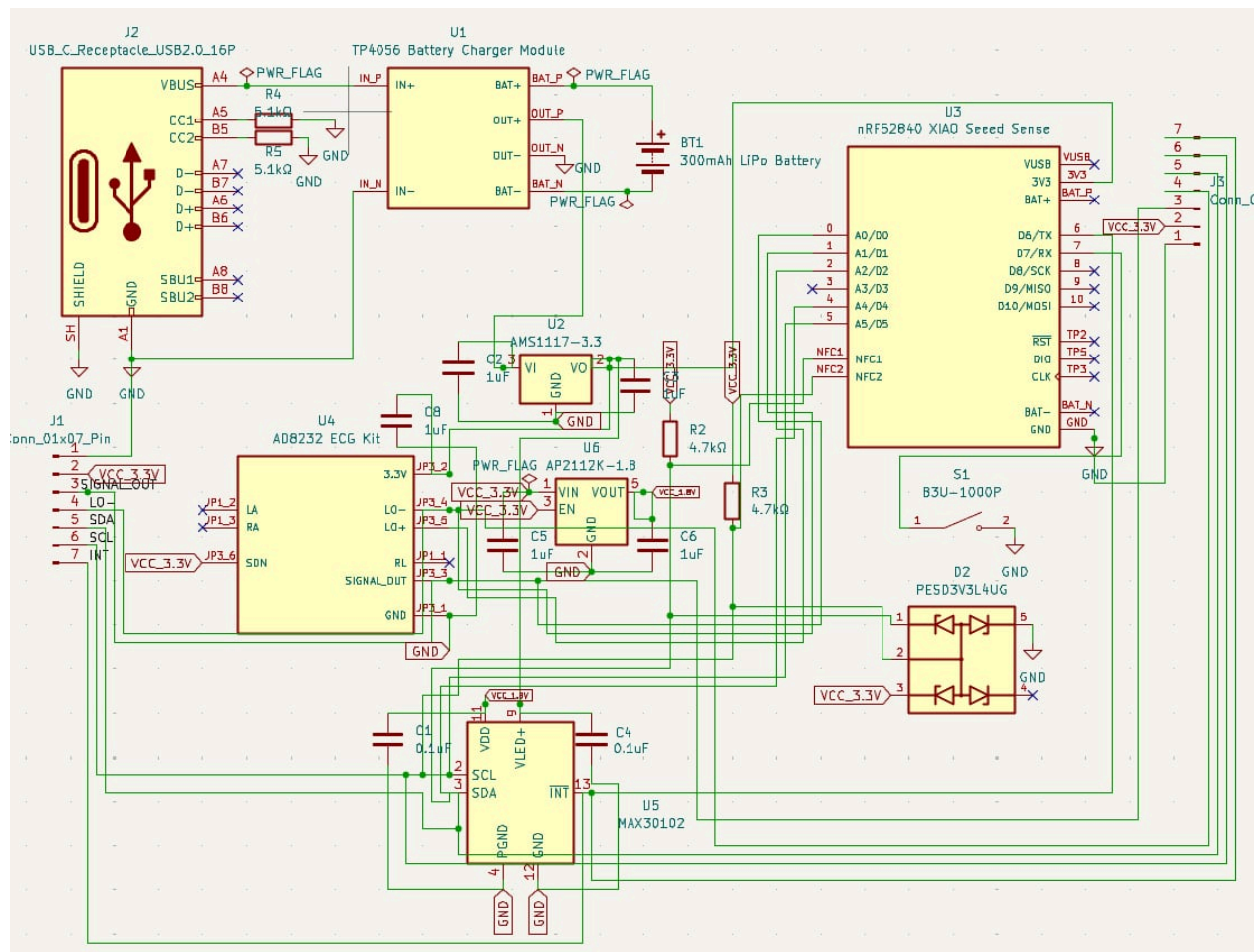


Figure 3

Proposed Schematic Diagram for the TIBOK Device Printed Circuit Board.

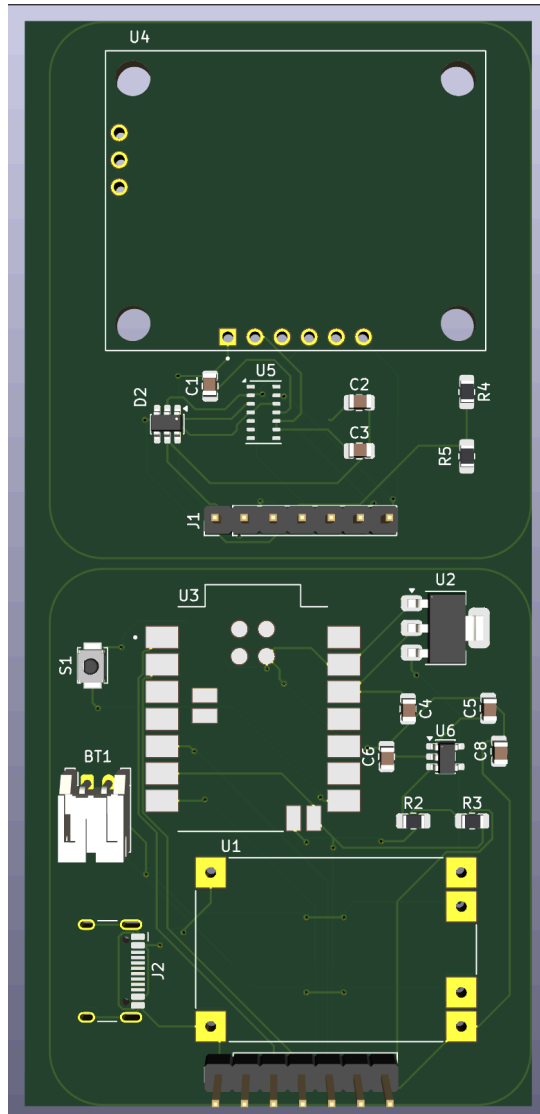


Figure 4

Proposed Printed Circuit Board Layout for the TIBOK Device.

MODEL ARCHITECTURE

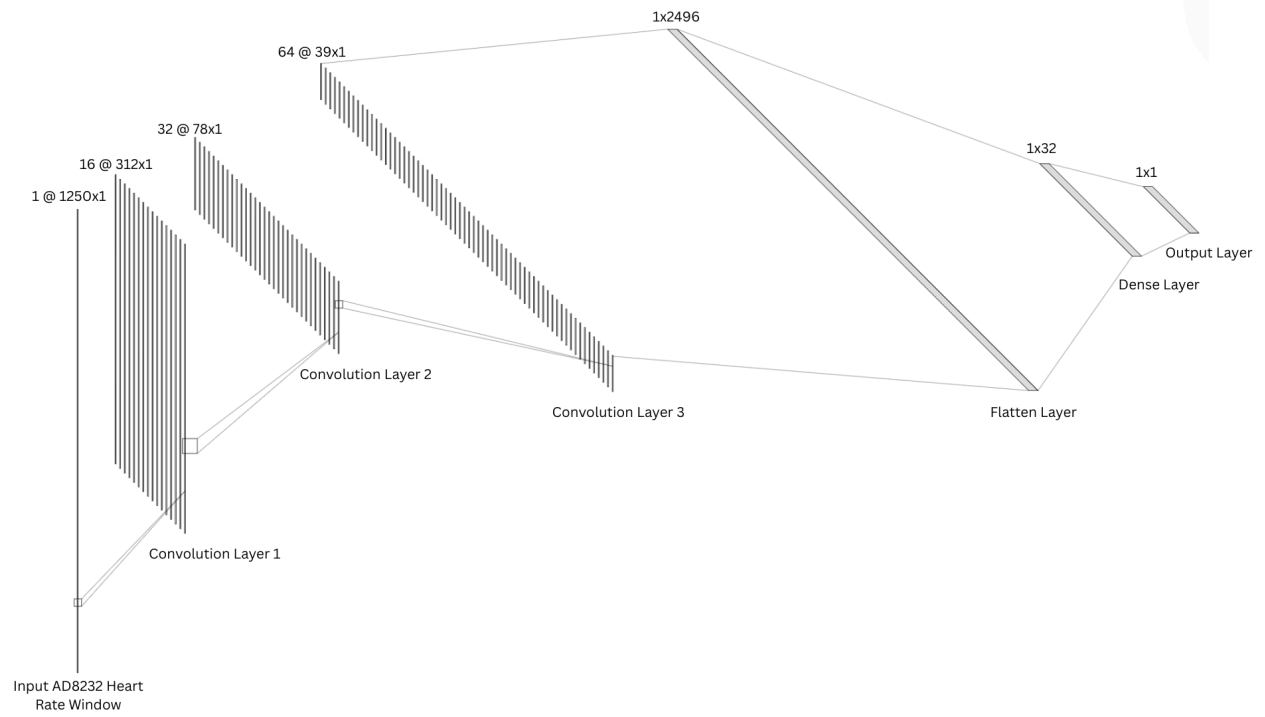


Figure 5

Proposed 1D-CNN Model Architecture for TIBOK Framework.

DATA PIPELINE

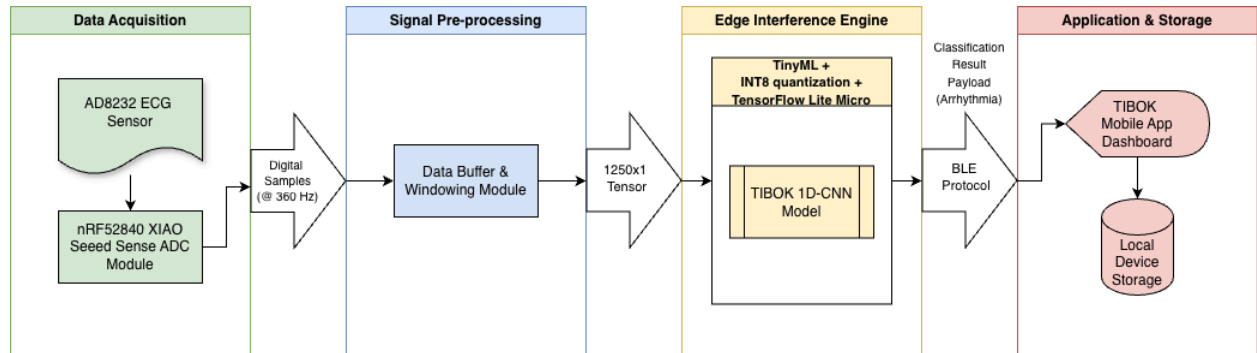


Figure 6

End-to-End System Architecture and Data Pipeline for TIBOK Framework from Data Acquisition to Application.

USER INTERFACE DESIGN



Figure 7.1

Home Screen (Active / Stop Monitoring State).

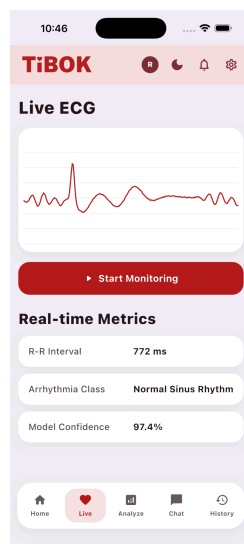


Figure 7.2

Live ECG / Real-Time Vitals Monitoring Screen.

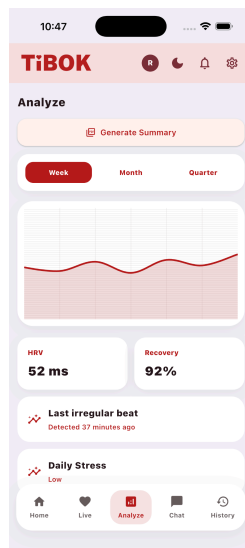


Figure 7.3
Heart Analysis / Signal Waveform Analytics Screen.

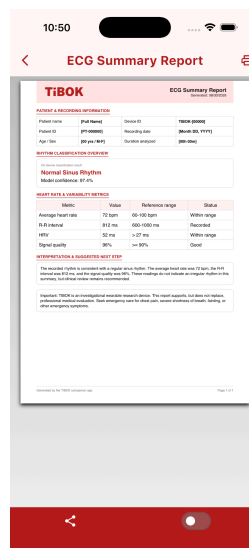


Figure 7.4
ECG Summary Report.

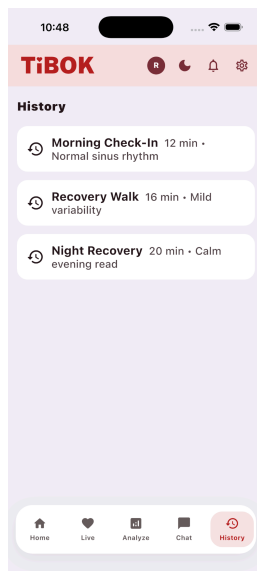


Figure 7.5

Session History.

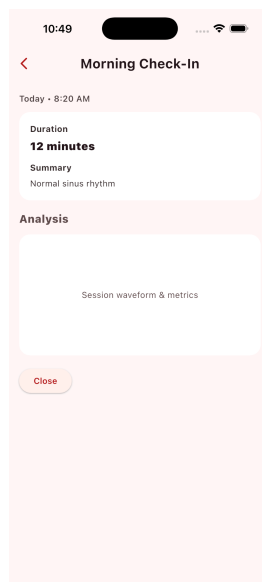


Figure 7.6

Session History.

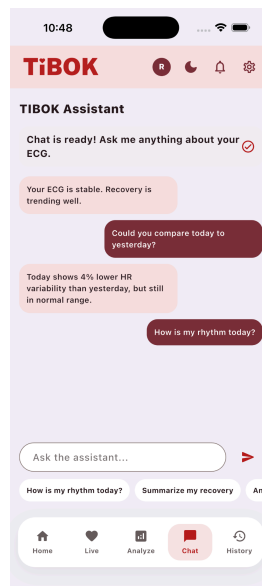


Figure 7.7

AI Chat Bot / Assistant

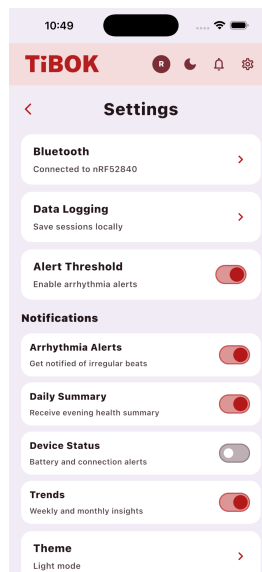


Figure 7.8

Settings / Menu

B. HARDWARE: SCHEMATIC DESIGN & PCB PROTOTYPING

To establish a low-noise, wearable electronic system, a custom multi-layer printed circuit board (PCB) layout will be engineered using electronic design automation tools. This process will focus on optimizing trace routing to isolate sensitive analog paths from digital high-frequency switching lines. According to Goyal et al. (2022), maintaining strict spatial separation of at least 1 mm between high-frequency digital clocks and analog input stages is paramount to prevent parasitic capacitive coupling and trace-to-trace crosstalk. By meticulously managing trace geometry, power plane distributions, and ground loops at this initial phase, the hardware architecture will be hardened against high-frequency board-level noise before physical component fabrication occurs. Following schematic finalization, the PCB will be physically fabricated using the toner-transfer method, wherein the printed circuit pattern is thermally transferred onto the copper-clad laminate board using a heat gun, followed by immersion in ferric chloride (FeCl_3) solution to etch away exposed copper. The etched board will then be rinsed, dried, and drilled using the PCB drill bit set to create through-holes for component leads, after which the board will be cleaned with isopropyl alcohol to remove residual flux and etching residue prior to soldering.

C. SOFTWARE: MODEL ARCHITECTURE TRAINING

Concurrently with the hardware schematic design, the deep learning pipeline will initiate by building and training the core diagnostic network within a high-precision desktop development environment in Google Colab. This step will utilize standardized, clinically annotated time-series records to teach the network to identify complex cardiac morphologies. As demonstrated by Ahmed et al. (2023), training a One-Dimensional Convolutional Neural

Network (1D-CNN) directly on raw, amplitude-normalized mathematical arrays from the MIT-BIH Arrhythmia Database will yield superior feature extraction capabilities for morphological anomalies compared to traditional manual heuristic modeling. Specifically, the annotated MIT-BIH records will be partitioned at the patient level into training (70%), validation (15%), and test (15%) subsets to prevent data leakage, and the network will be trained using the Adam optimizer with a categorical cross-entropy loss function across a fixed number of epochs, with early stopping applied based on validation loss to prevent overfitting. This desktop-bound training phase will establish the optimal baseline diagnostic accuracy, sensitivity, and specificity parameters necessary for future performance benchmarking.

D. HARDWARE: ENCLOSURE FABRICATION

Once the structural dimensions of the custom PCB layout are locked, the physical form factor of the device will be developed by designing an ergonomic, wrist-worn smartwatch chassis. The enclosure will be modeled using computer-aided design (CAD) software and realized using PETG 3D printer filament with a Creality Ender-3 V3 SE 3D printer. This phase will be critical to ensure structural rigidity, secure battery containment, and proper mechanical orientation for the physical sensors. Ha et al. (2026) emphasize that the structural design of portable, clinical-grade telehealth wearables must prioritize compact, rigid component stabilization to safeguard internal connections from mechanical strain during everyday user movement.

E. SOFTWARE: UI/UX DESIGNING

Simultaneously running alongside enclosure development, the frontend client application will be constructed using a cross-platform software framework in Visual Studio Code version

1.132 using the Flutter version 3.44.9 UI software development toolkit with the Dart version 3.12.2 programming language to establish an interactive data visualization ecosystem. This process will involve programming the application logic to map raw real-time arrays into geometric tracking dashboards. The integration of time-series tachograms and non-linear Poincaré scatter plots within the user interface will significantly enhance diagnostic readability. Research by Gruwez et al. (2021) in the INTERPRET-AF study indicates that supplementing basic heart rate data with geometric tachograms and Poincaré point distributions will provide clinicians with distinct visual signatures, maximizing diagnostic visibility for transient cardiac events over standard text readouts alone.

F. HARDWARE: POWER & ASSEMBLY

This step will focus on assembling the physical hardware and stabilizing the internal power distribution networks. Components will be soldered onto the custom PCB prototype, and nRF52840 XIAO Seeed Sense microcontroller's dedicated power management will be configured alongside low-dropout (LDO) linear regulators and the TP4056 battery charging module to step down the lithium-polymer battery voltage to a highly consistent 3.3V operating rail. Prior to full system integration, the AD8232 ECG module and the MAX30102 pulse oximeter will each be individually bench-tested by connecting them directly to the nRF52840's analog and I²C pins and verifying their raw output against a known reference signal, ensuring that each sensor performs within expected parameters before being combined into the complete sensing pipeline. Maintaining a clean power rail will be essential, as voltage fluctuations from the power source can introduce low-frequency ripple artifacts. Chandani et al. (2025) noted that incorporating aggressive voltage regulation and hardware decoupling capacitors is mandatory in

low-cost ambulatory monitors to prevent systemic power-supply noise from corrupting delicate biopotential sensor readouts.

G. SOFTWARE: MODEL QUANTIZATION & TESTING

In parallel with physical board assembly, the heavy, high-precision desktop neural network will undergo structural optimization to bridge the gap between desktop computing and embedded processing. Utilizing post-training INT8 full integer quantization in Google Colab, the 32-bit floating-point (FP32) network weights will be compressed into an optimized 8-bit integer (INT8) format, generating a highly efficient flatbuffer file. As established by Gupta and Shivhare (2025), applying INT8 quantization through TinyML framework pipelines will allow deep learning networks to achieve high diagnostic accuracy while drastically reducing the model's overall computational footprint and memory requirements, making execution feasible on resource-constrained microcontrollers.

H. HARDWARE: NOISE FILTERING & DIGITAL SIGNAL PROCESSING

The firmware will be developed and flashed onto the nRF52840 XIAO Sseed Sense using the Arduino IDE with the Sseed nRF52 board package, allowing the research team to program the microcontroller's on-chip DSP routines directly. With the hardware assembled, the local microcontroller firmware will be programmed to execute on-chip digital signal processing (DSP) to clean the raw analog bio-signals immediately upon ingestion. It will use a 0.1 μ F ceramic capacitor to reduce the signal noise. This pipeline will apply a digital band-pass filter (0.5 Hz to 40 Hz) to remove baseline wander and muscle artifacts, paired with a digital notch filter configured to eliminate localized ambient power-line hum. Kim et al. (2023) demonstrate that executing digital signal filtering locally on an embedded edge-node, rather than offloading

raw, noisy signals to a remote server, drastically reduces the system's power consumption and prevents wireless data transmission bottlenecks, proving that on-chip DSP will be essential for continuous monitoring.

I. SOFTWARE: BLUETOOTH LOW ENERGY DATA SERIALIZATION

Operating concurrently with the on-chip DSP development, the wireless telemetry stack of the system will be built to manage data packet structure and transmission protocols. This phase will configure the communication firmware to compress calculated R-R intervals, heart rate metrics, and model arrhythmia classification state flags into tightly structured, serialized byte arrays. Sekar et al. (2026) state that utilizing optimized Bluetooth Low Energy (BLE) Generic Attribute Profile (GATT) communication channels will allow low-cost edge prototypes to maintain highly reliable wireless telemetry links with companion devices, ensuring continuous data visualization without draining the wearable's limited battery reserves.

J. INFERENCE LOOP OPTIMIZATION

Prior to any procedure involving human participants, the study protocol will be submitted to the school's research ethics review committee for clearance, and only participants who provide informed consent after being briefed on the study's purpose, procedures, and risks will be enrolled. The finalized, integrated smartwatch prototype will be moved into a dedicated field-testing protocol consisting of five independent evaluation trials using randomized data blocks from the MIT-BIH Arrhythmia Database (Flores-Castañeda et al., 2025).

To evaluate diagnostic accuracy and efficacy, the system will measure the classification F1-score and specificity to determine how accurately it will differentiate between normal sinus rhythms and ventricular or supraventricular arrhythmias. This will be operationalized by

recording the classification outputs generated by the local INT8 model on the mobile client app during live streaming, which will then be mapped into a 2 x 2 confusion matrix against the certified clinical annotations provided by the MIT-BIH dataset (Ahmed et al., 2023).

On-chip processing latency will also be documented to assess execution speed, specifically capturing the precise timeframe required for the system to execute local digital filtering and a single 1D-CNN deep learning inference pass. This will be measured via embedded microsecond timers [`micros()`] programmed into the nRF52840 firmware that calculate the exact time differential from the moment an ECG window array is completely buffered to the final execution step of the neural network prediction layer.

Active electrical power consumption will be trackable to evaluate wearable sustainability, monitoring the real-time electrical current drawn by the system during baseline operations, active signal processing, and continuous BLE data streaming. This measurement will be achieved by connecting an inline, high-resolution hardware power analyzer or digital multimeter directly to the positive battery rail of the smartwatch prototype to capture the continuous average milliampere (*mA*) draw over each 30-minute test run (Chandani et al., 2025).

Finally, wireless telemetry packet loss will be calculated to gauge the reliability and stability of the BLE GATT communication link between the smartwatch chassis and the mobile application client. This parameter will be monitored by appending a rolling transaction packet index counter to the serialized byte streams within the firmware, which will allow the Flutter application to parse these sequential header IDs and compute the exact ratio of dropped or corrupted data packets over the total number of packets transmitted during the trial (Sekar et al., 2026).

X. SAMPLING SIZE AND SAMPLING METHOD

The study will use purposive sampling, a non-probability sampling technique in which participants will be deliberately selected according to predefined inclusion criteria aligned with the objectives of the study. This sampling approach will be appropriate because the investigation will require two clearly defined groups for comparison, namely participants with a history of arrhythmia and participants without a history of arrhythmia. Purposive sampling will therefore allow the selection of information-rich cases that are directly relevant to the research problem. (Memon et al., 2025; Magakwe et al., 2025).

The study will target a total of 10 participants, composed of 5 participants with a history of arrhythmia and 5 participants without a history of arrhythmia. This sample size will be considered suitable for a pilot validation study because pilot and feasibility studies are primarily intended to assess practicality, preliminary performance, and methodological procedures. Current guidance on pilot studies indicates that formal power calculations are not always required; instead, the sample size should be justified in relation to the specific objectives of the study, the precision of the estimates required, and the available resources (In et al., 2024; Kunselman, 2024).

The selected sample size will also be consistent with recommendations that pilot studies should be large enough to generate useful preliminary estimates for outcomes such as agreement, sensitivity, specificity, and overall device performance, while remaining feasible within the constraints of an exploratory prototype study. In addition, the division of participants into two purposively selected groups will ensure that both comparison conditions are represented, thereby

strengthening the preliminary evaluation of the wearable ECG prototype against the reference method (In et al., 2024; Montgomery, 2025).

Overall, the sampling method and sample size will be aligned with the exploratory and feasibility-oriented nature of the study. The use of purposive sampling and a total sample size of 10 participants will provide a practical and defensible basis for the preliminary validation of the TIBOK system.

XI. DATA COLLECTION

The empirical dataset for this study will be systematically compiled through a digital extraction pipeline using established, open-access cardiac repositories. Telemetry inputs will be sourced directly from the MIT-BIH Arrhythmia Database via the PhysioNet atmospheric research platform. The collection process will target an evenly distributed selection of continuous two-channel ambulatory ECG recordings, focusing specifically on leads that isolate prominent R-wave vectors, such as modified limb lead II (MLII) and chest lead V1. The raw time-series data, originally sampled at 360 Hz with 11-bit resolution over a ± 10 mV range, will be downloaded alongside their corresponding gold-standard reference annotation files (.atr). These annotations, manually verified by independent clinical cardiologists, categorize every individual heartbeat complex into distinct diagnostic classes, including Normal Sinus Rhythm (NSR) and arrhythmia. To prevent data leakage between model training and evaluation, patient records will be partitioned at the patient level prior to any preprocessing, such that no single patient's beats appear in both the training and evaluation sets. The extracted signal blocks will then be locally cached in a structured directory format, creating a standardized, immutable baseline dataset that eliminates any external telemetry drift before it enters the digital signal filtering and

preprocessing pipelines.

For human testing, the sample size will be 10 participants, consisting of 5 individuals with a history of arrhythmia and 5 individuals without a history of arrhythmia. For participants whose arrhythmia presents intermittently rather than continuously, multiple or extended recording sessions will be conducted to increase the likelihood of capturing an arrhythmic episode within the observation window. In addition, participants will be selected through purposive sampling based on the inclusion criteria established for the study. Our participants may include individuals referred for ECG evaluation as well as those with existing or known suspected rhythm abnormalities. For the individuals with unusable recordings, excessive motion artifacts, or incomplete data, their data will not be included for the final analysis. For how we'll get the data, each participant will undergo ECG recording using both the TIBOK device and the GE MAC 5500 system during the same session. The wearable device and the MAC 5500 will be attached accordingly. The recordings will be performed as simultaneously as possible so that both devices capture the same cardiac condition during the same window. The GE MAC 5500 tracing will serve as the reference record, but the final ground-truth label will be assigned by our qualified cardiologist who is blinded to the wearable device. Our cardiologist will then classify each case as arrhythmia or non-arrhythmia, and further specify the arrhythmia sub-type is applicable. By blinding the cardiologist, we reduce bias and match validation approaches. After the ECGs are acquired, the TIBOK output will be converted into the same binary or categorical label used for the GE MAC 5500. Any unreadable or low-quality traces will be recorded separately and will be excluded from the primary accuracy analysis. For the primary outcome, it will be the diagnostic agreement between our device, TIBOK, and the MAC 5500 reference

label. For the secondary outcomes, it will include sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), accuracy, and Cohen's kappa (κ).

XII. DATA ANALYSIS

The statistical evaluation of the framework will follow a quantitative, comparative design to measure the diagnostic precision and computational efficiency of both algorithmic tracks. Upon processing the synchronized 5-second matrix segments, the generated output arrays from the pipeline and the 8-bit quantized 1D-CNN model will be evaluated against the clinician-verified reference annotations. Model training and evaluation blocks will be drawn from non-overlapping patient records to ensure that reported performance metrics reflect generalization to unseen individuals rather than memorization of patient-specific signal characteristics. Performance will be mathematically broken down using standard binary and multi-class confusion matrices to track true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). From these values, the diagnostic capability will be quantified by calculating the overall system accuracy

$$\text{Overall System Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\%$$

, which measures the total percentage of true positive and true negative arrhythmia assessments for each track.

To deeply evaluate the system accuracy level, the overall percentage will be contextualized and explained by breaking down the rate of diagnostic errors into two distinct performance metrics:

the Type I Error or False Positive Rate $\text{Type I Error Rate (FPR)} = \frac{FP}{TN + FP} \times 100\%$, which

measures the rate at which the integrated system falsely visualizes and detects an arrhythmia when the rhythm is actually normal, and the Type II Error or False Negative Rate

$$\text{Type II Error Rate (FNR)} = \frac{FN}{TP + FN} \times 100\%$$

which measures the rate at which the integrated

system fails to detect and visualize an actual arrhythmia event. Furthermore, the framework's operational capacity will be assessed by calculating the specific diagnostic performance and

timing characteristics via specificity

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100\%$$

, F1-score

$$\text{F1-score} = \frac{2 \times TP}{2 \times TP + FP + FN} \times 100\%$$

and latency

$$t_{\text{latency}} = t_{\text{output}} - t_{\text{input}}$$

. To

determine whether the data-driven TinyML model yields a statistically significant improvement over the deterministic, classical signal processing algorithm, the compiled performance metrics will be subjected to paired-sample t-tests or Wilcoxon signed-rank tests, depending on the normality distribution of the cross-validation scores. Resource efficiency profiles, specifically latency measured via microsecond tracking loops and dynamic SRAM heap usage, will be cross-referenced with the diagnostic scores using a joint resource-to-accuracy ratio analysis. This statistical correlation will isolate the exact trade-offs between mathematical precision and edge hardware footprint, providing the empirical justification required to determine the optimal processing topology for resource-constrained medical devices. For the human testing analysis, a confusion matrix will be constructed from the wearable prediction and cardiologist reference classification. From this table or matrix, sensitivity, specificity, PPV, NPV, and accuracy will be calculated. Given the small pilot sample size (n=10), sensitivity, specificity, PPV, and NPV will be reported with 95% confidence intervals calculated using the Wilson score method, which is

more reliable than the standard normal approximation for small samples. Because the human validation sample was recruited at an approximately 1:1 ratio of participants with and without arrhythmia history, rather than reflecting population-level prevalence, the resulting PPV and NPV are specific to this enriched sample and should not be interpreted as real-world predictive values. To address this, a secondary sensitivity analysis will apply Bayes' theorem to the observed sensitivity and specificity across a range of regional prevalence estimates rather than a single fixed value: 0.5% (low, Malaysia), 3.2% (mid, drawn from a general-population screening study using a comparable single-lead ECG methodology; Yuniadi et al., 2024), and 5.4% (high, reflecting older or higher-risk adults; Chan et al., 2022). This range was selected given the continued scarcity of Philippine-specific arrhythmia prevalence data (Ceriales et al., 2022), and provides a more realistic picture of TIBOK's real-world predictive value across differing population contexts than a single assumed prevalence would. Using the measured sensitivity and specificity of TIBOK, the prevalence-adjusted predictive values will be computed as follows:

$$PPV = \frac{\text{sensitivity} \times \text{prevalence}}{\text{sensitivity} \times \text{prevalence} + (1 - \text{specificity})} \quad NPV = \frac{\text{sensitivity} \times \text{prevalence}}{\text{sensitivity} \times \text{prevalence} + (1 - \text{specificity})} \cdot \quad \text{This}$$

secondary analysis will provide a more clinically realistic estimate of the device's screening performance in a general adult population, where the prevalence of arrhythmia is expected to be lower than in the validation sample. Sensitivity and specificity are unaffected by this design, as

they are prevalence-independent. The Cohen's kappa (κ , confidence interval) $\kappa = \frac{p_0 - p_e}{1 - p_e}$,

measures the level of agreement between TIBOK and the cardiologist-interpreted GE MAC 5500 12-lead ECG findings in classifying arrhythmia versus non-arrhythmia. For the results, less than 0: No agreement (worse than random) 0.01 - 0.20: None to slight, 0.21-0.40: Fair agreement,

0.41 - 0.60: Moderate Agreement, 0.61 - 0.80: Substantial Agreement, 0.81-1.00: Almost perfect agreement.

SCOPES AND LIMITATIONS

The TIBOK hardware framework and system architecture will encompass an embedded edge-computing device using the nRF52840 XIAO Seeed Sense microcontroller along with an AD8232 single-lead biopotential electrocardiogram (ECG) analog front-end and a MAX30102 optical pulse oximetry sensor module. User input and hardware safety will be accounted for by an Omron B3U-1000P event-marker tactile button, which will be used to enable instant symptom logging during monitoring, along with a Nexperia PESD3V3L4UG quad-channel TVS diode array to provide electrostatic discharge protection across signal lines. The system will be integrated into a custom printed circuit board (PCB) housed within a wrist-worn 44 mm by 28 mm custom 3D-printed chassis featuring a 4 mm photoplethysmography (PPG) sensor window in between two 6 mm ECG conductive pads. Telemetry relies on Bluetooth Low Energy (BLE 5.4) protocols to stream continuous ECG and PPG data to a dedicated cross-platform Flutter mobile dashboard. The study is limited to one prototype unit (n=1), excluding device-to-device manufacturing variation and production-scale reliability. The system hardware is strictly delimited from multi-lead or 12-lead diagnostic ECG arrays and cloud-dependent processing architectures or continuous server-side model retraining.

For on-device signal processing and evaluation, the software pipeline employs a localized 1D Convolutional Neural Network (1D-CNN) optimized through 8-bit integer quantization (INT8) and executed via TensorFlow Lite for Microcontrollers. Model validation is conducted in-silico by benchmarking the quantized embedded network against a standard 32-bit

floating-point (FP32) baseline using annotated recordings from the open-access MIT-BIH Arrhythmia Database. Performance is quantified strictly across inference latency, memory footprint, F1-score, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and Cohen's kappa (κ). The evaluation excludes heavy continuous cloud-based AI processing, online model retraining, spatial environmental mapping, multi-modal vision models, and automated multi-disease or arrhythmia-subtype classification. TIBOK is limited to the predefined classification task of distinguishing target arrhythmia pattern(s) from non-arrhythmic recordings, while cardiologist-identified subtypes serve only as reference annotations.

Operationally, the system is designed as a low-power, point-of-care cardiac screening and assistive monitoring tool tailored for resource-constrained primary care settings. Biopotential signal acquisition is bounded by single-channel filtering equivalent to a Modified Limb Lead II setup. Participant testing, where applicable, will be limited to adults aged 18–65 years recruited from the designated study site in Quezon City, excluding pregnant individuals and participants with pacemakers, implantable cardioverter-defibrillators (ICDs), or other implanted cardiac devices. Findings are therefore limited to the recruited population and single-site setting and may not generalize to other Philippine regions or demographic groups. The project is explicitly delimited from acting as a certified standalone medical diagnostic device, replacing hospital-grade 12-lead clinical ECG monitors, or undergoing direct clinical trial interventions and multi-center human patient validations. Although IEC 60601 standards may serve as engineering references, formal certification and regulatory conformity assessment are outside the study.

Furthermore, operational reliability is limited to controlled indoor and point-of-care environments. The prototype will not be evaluated for water resistance, immersion, bathing,

swimming, or prolonged moisture exposure, and no specific IP rating or broad environmental operating range will be claimed. Excessive perspiration, high-motion outdoor activities, rough terrain usage, and electromagnetic interference exceeding single-lead analog filtering capabilities are outside the scope. Continuous-wear evaluation is likewise limited to the experimentally verified battery operating duration and does not replicate the 24–48-hour monitoring duration of conventional Holter systems. Battery endurance beyond the verified operating period, long-term uninterrupted wear, and charging performance are excluded from the study.

RISK AND SAFETY

This research encompasses building and evaluating an initial prototype of the TIBOK, a smart Edge-IoMT wearable wristband that will monitor cardiovascular activity for research and educational purposes. The project consists of Printed Circuit Board (PCB) fabrication, assembling the wearable hardware, developing embedded software, lab tests, and a 50-participant validation trial. The study will implement safety measures to mitigate electrical, chemical, thermal, biological, mechanical, ethical, and cybersecurity risks. The prototype is a research-grade model and is not intended to substitute professional cardiac medical devices or provide a valid medical diagnosis.

There are electrical safety concerns associated with assembling circuits and testing the prototype using a 3.7 V 300 mAh LiPo battery. The latter can be damaged, ruptured, and shorted, leading to overheating, electrolyte leakage, and combustion under conditions of physical damage, overcharging, short-circuiting, and high temperatures. All hardware assembling and maintenance procedures will be performed after switching off the main power supply. The TP4056 module with built-in protection circuitry will be used for LiPo battery charging. A digital multimeter will be used to check the continuity of the circuit before powering up the prototype. Finally, all other safety measures should be taken when working with the battery, including proper storage and transportation, to avoid mechanical damage and short-circuiting (Chandani et al., 2025; Armand et al., 2023).

PCB fabrication involves corrosive ferric chloride solution (FeCl_3), while soldering poses inhalation hazards from solder fumes. FeCl_3 can cause skin irritation and chemical burns if the solution is accidentally transferred to the skin (Hill Brothers Chemical Company, 2026). Nitrile gloves, laboratory coats, and chemical splash goggles must be worn when handling FeCl_3 .

It is critical to ensure adequate ventilation in the laboratory and dispose of chemical waste according to established institutional regulations. A fume extractor will be used to suck solder fumes away while performing soldering tasks, and nitrile gloves and safety glasses will be worn while handling hot soldering tools and molten solder (Arshon, n.d. , 2025).

Mechanical and thermal hazards can arise from working with soldering irons, heat guns, drill machines, and cutters. Such tools have a potential to cause thermal burns, lacerations, and punctures if not operated correctly. Safety glasses, heat-resistant mats and soldering stands, and electrostatic discharge (ESD) safety precautions will be employed to minimize the risk of mechanical and thermal injuries during fabrication (Arshon, n.d., 2025).

The study involves minimal biological safety risks because the prototype is non-invasive and only acquires medical-grade ECG readings. Disposable Ag/AgCl electrodes and MAX30102 optical heart sensors will be used to record heart activity and saturation levels, respectively. Prior to applying ECG electrodes, the participants' skin will be cleaned with alcohol wipes to promote better adhesion of the electrodes and reduce the risk of skin irritation caused by impaired skin-electrode impedance (Goyal et al., 2022). Correct electrode placement techniques will also be used to optimize ECG recording and minimize placement-related artifacts (GE HealthCare, 2025). The study will exclude individuals with open wounds, skin ulcers, and allergies to adhesive materials. Disposable ECG electrodes will be used for every participant, while the prototype's surface will be sanitized prior to each procedure (Fabian & Ahmed, 2026).

The MAX30102 optical sensor is generally safe in terms of exposing the skin to low-level light to calculate heart rate and peripheral oxygen saturation. However, incorrect placement or movement of the prototype can lead to inaccurate readings. The participants will be educated on the proper placement of the device to ensure optimal performance. In addition, the

prototype will be adjusted in a manner that it does not cause discomfort on the wrist. If a participant feels uncomfortable, they have an option to remove the prototype or request to discontinue the procedure (Filgueiras et al., 2025; Oberdier et al., 2023).

Participants will be prioritized throughout the research process. They will be informed about the study's purpose, protocols, potential benefits, and risks before providing informed consent to participate voluntarily. They can also withdraw their consent or discontinue participation at any moment without negative consequences. ECG results from TIBOK will be compared to standard-of-care 12-lead ECG recordings taken with GE MAC 5500 and interpreted by a licensed cardiologist. The prototype's predicted results will only be used for research and educational purposes and should not be used as a diagnostic tool (Chung et al., 2022; Siontis et al., 2021).

As the prototype uses Bluetooth Low Energy (BLE) to transmit physiological data to the cloud, the study will prioritize participants' privacy and data confidentiality. Physiological data will be de-identified and encrypted during transmission and storage. All data will be stored in secure cloud servers protected by complex passwords accessible only to designated researchers authorized by the study's principal investigator. Similarly, participants' personal information will be stored separately from physiological data to ensure confidentiality and privacy (Armand et al., 2023; Serrani & Aliverti, 2024).

The research involves potential technical limitations such as intermittent BLE connections, motion artifacts, electrode disconnection, electrical noise, and internet connectivity issues. Such limitations may compromise data acquisition integrity and impede continuous data transmission to the cloud. To minimize these limitations, the mobile application will save all data in memory during communication disruptions and automatically sync them when connectivity is

re-established. In addition, signal pre-processing and digital filtering will be applied to ECG data to enhance its quality before using it for AI model training procedures (Safdar et al., 2024; Serrani & Aliverti, 2024).

Irregular heart rhythms identified by the prototype should be regarded as screening results only and should not be used as primary diagnostic indicators. AI-assisted diagnostic tools should be considered as complementary measures to conventional 12-lead ECG and should not replace professional medical advice. The results of ECG recorded by the prototype must be ratified by standard-of-care cardiac screenings before making any medical treatment decisions or issuing diagnosis statements (Afridi et al., 2024; Martínez-Sellés & Marina-Breyse, 2023; Siontis et al., 2021).

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PLAGIARISM CHECKER RESULTS

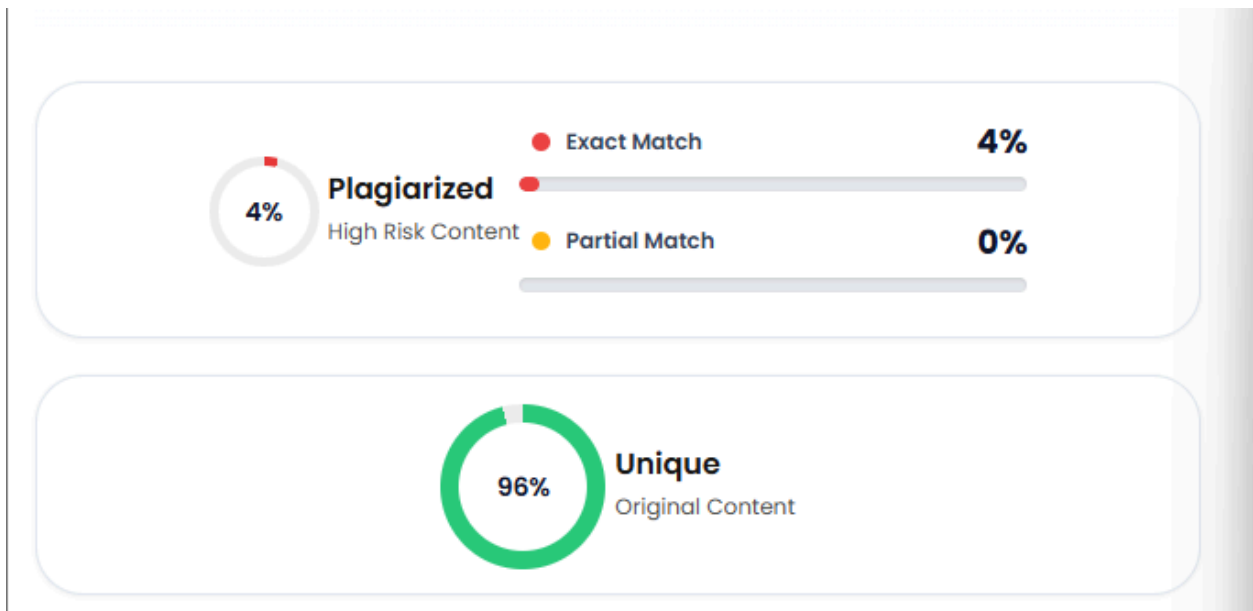
The manuscript's originality was verified using Duplichecker. The overall similarity index was 4%, which is well within the acceptable range for academic research papers. This result indicates that no instances of plagiarism were identified, and the text consists of original content.

Tool Used: Duplichecker

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AI Declaration

The researchers used Perplexity AI, OpenAI's ChatGPT, and Google Gemini to assist in refining the grammar of the introduction and to generate possible formulations of research questions. No AI tool was used to generate results. All final content was verified and authored by the researchers.

Prompts

1. Revise this introduction to improve grammar and academic tone without changing the meaning.
2. Make this paragraph more concise while retaining all key information and citations.
3. Improve the clarity and flow of this rationale for a science research paper.
4. Rewrite this sentence to sound more formal and suitable for academic writing.
5. Check this paragraph for grammatical errors and suggest corrections.
6. Paraphrase this section to avoid redundancy while preserving technical accuracy.
7. Suggest stronger transition words to improve coherence between these paragraphs.
8. Classify these research questions as descriptive, comparative, or relational.
9. Refine these research questions to make them more precise and measurable.
10. Convert these objectives into testable hypotheses (null and alternative).
11. Improve the wording of these hypotheses to follow standard scientific format.
12. Identify the dependent variables from these research objectives.
13. Suggest clearer definitions for these variables in an experimental context.
14. Rewrite this methodology description to improve clarity and structure.
15. Check if the tone of this research paper is consistent and formal throughout.
16. Suggest ways to make this paragraph more concise and less repetitive.

17. Reword this section to enhance readability for a scientific audience.
18. Review this text and suggest minor edits for grammar, clarity, and coherence only.